INCH-POUND

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MILITARY SPECIFICATION

CONTAINER, SHIPPING AND STORAGE, MK 300 MOD 0 AND MOD 1 (FOR TORPEDO MK 46 MOD 0 THROUGH MOD 5)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Shipping and Storage Container Mark 300 Mod 0 and Mod 1 for Torpedo Mk 46 Mod 0 through Mod 5, herein referred to as the "container" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-15024	Plate, Tags and Bands for Identification of Equipment

STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging and Packing Materials, Test Procedures

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 3730), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 8140

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MILITARY

MlL-STD-129	Marking for Shipment and Storage
MIL-STD-648	Design Criteria for Specialized Shipping Containers
MlL-STD-831	Test Reports, Preparation of

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2. Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

DL2613831	Container, Shipping and Storage Mark 300 Mod 1
LD541201	Container, Shipping and Storage Mark 300 Mod 0
2452719	Breather Valve

(Application for copies of NAVSEA drawings should be addressed to Commanding Officer, Naval Ordance Station, Crane Division, Naval Surface Warfare Center, Louisville Kentucky, 40214-5001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

PUBLICATIONS

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that

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prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable detail specifications (see 6.2). In the event of any conflict between the requirements of this specification and the associated detail specification, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.5.

3.3 Materials and treatments of materials and parts. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the container shall be in accordance with the requirements specified herein, and with applicable documents listed on LD541201 and DL2613831. No fungus nutrient shall be used.

3.4 Design and construction. The container shall be of the design, construction, and physical dimensions specified on the drawings and specifications listed on LD541201 and DL2613831.

3.4.1 Gauges. The contractor shall provide whatever gauges are necessary and adequate to ensure that the material and parts of the test set will meet the dimensions and requirements shown on applicable documents. When requested by the contractor, the contracting activity will furnish existing pertinent Navy drawing of final acceptance gauges for use only in designing of the contractor's gauges. This action does not relieve the contractor of the responsibility for designing and manufacturing the gauges needed for satisfactory fulfillment of the contract requirements.

3.4.2 Inspection and test equipment and facilities. The contractor shall furnish and maintain all necessary inspection and test equipment and facilities and provide personnel for performing all quality conformance tests. The test equipment shall be adequate to permit performance of the required quality conformance tests for the container.

3.5 Performance requirement.

3.5.1 Airtightness. The container shall not leak air at a rate greater than 0.3 pounds per square inch gauge (psig) when tested in accordance with 4.7.2.

3.6 Environmental requirements.

3.6.1 Vibration. The container, in its normal shipping attitude, with one Torpedo main assembly of gas generator, or simulated load (see 6.1), shall withstand vibration (see 6.5.2) test of 4.5.2, 4.5.2.1, and 4.5.2.2 and thereafter show no evidence of damage to container or contents and meet the performance requirement of this specification.

3.6.1.1 Variable frequency. The container shall withstand vibration of 5 to 60 Hertz (Hz) when tested

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in accordance with 4.5.2.1.

3.6.1.2 Natural frequency. The container shall withstand vibration at the resonant frequency of the packaged-article cushioning system when tested in accordance 4.5.2.2.

3.6.2 Rough handling. The container, with one Torpedo main assembly or gas generator, or simulated load enclosed, shall withstand the test of 4.5.3 and thereafter show no evidence of structural damage to container or contents or shifting of contents.

3.6.2.1 Cornerwise-drop (rotational) test. The container shall withstand shock loading as induced by a cornerwise drop of not less than 24 inches on a steel or concrete surface on each of the 4 base corners when tested in accordance with 4.5.3.1.

3.6.2.2 Impact test. The container shall withstand impact on all four vertical surfaces against a rugged unyielding surface at a velocity of 7.6 feet per second when tested in accordance with 4.5.3.2.

3.6.2.3 Tip-over test. The container shall withstand being tipped slowly from its base to fall freely on each side on a steel or concrete surface when tested in accordance with 4.5.3.3.

3.6.2.4 Skids test. The container shall withstand being pushed or towed a distance not less than 5 feet in line with longitudinal axis and not less than 5 feet normal to the longitudinal axis when tested in accordance with 4.5.3.4.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess material, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. Inspection of the container shall be classified as follows:

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a. First article inspection
   (1) First article sample (see 4.3.1)
   (2) First article tests (see 4.5)

b. Periodic production inspection
   (1) Periodic production sampling (see 4.3.2)
   (2) Periodic production tests (see 4.6)

c. Acceptance inspection
   (1) Acceptance sampling (4.3.3)
   (2) Acceptance tests and examinations (see 4.7)

4.3 Sampling.

4.3.1 First article sample. A first article sample of one container manufactured in accordance with 3.3, shall be subjected to the first article tests detailed in 4.5 at a facility designated by the contracting activity. Acceptance of the first article sample shall be based on no defects in the sample. Further production of the container by the contractor, prior to approval of the first article sample, shall be at the contractor's risk.

4.3.2 Periodic production sampling. Periodic production tests detailed in 4.6 shall be performed at a facility designated by the procuring activity. Samples shall consist of units that have satisfactorily passed the acceptance tests and examinations detailed in 4.7, selected as follows:

a. Step 1. Upon initiation of production, randomly select 1 unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to step 2.

b. Step 2. Randomly select 1 unit from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to step 3.

c. Step 3. Randomly select 1 unit from each group of 50 units. Continue with step 3 for the balance of the contract or until a defective unit is found.

4.3.2.1 Defective units. When a defective unit is found in any group the supplier shall:

a. Determine the cause of the defect

b. Determine if the defect is an isolated case

c. Take the necessary action to eliminate the defects from units on hand and, if directed by the contracting activity, from those units previously shipped.

d. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

e. Return to step 1 (4.3.2a) and proceed as indicated for inspection of units subsequently produced.

4.3.3 Acceptance sampling. The tests and examinations detailed in 4.7 shall be conducted on each container to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

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4.4 Test conditions. Unless otherwise specified, the container shall be subjected to tests under the following conditions:

a. Temperature: Ambient (20þ to 1 10þF)
b. Humidity, relative (see 6.5.1): Ambient, up to 95 percent
c. Altitude: Normal ground atmospheric pressure

4.5 First article tests. The first article sample container with the Torpedo Main Assembly of Gas Generator (see 6.1) or simulated contents of equal weight distribution enclosed, shall be tested in accordance with 4.5.2 through 4.5.3.4. With approval of the contracting activity, testing shall be at ambient temperature only (see 4.4). The first article sample shall be subjected to and satisfactorily pass the acceptance tests and examinations both prior to and after completion of the first article tests. The first article tested sample shall be plainly identified by securely attached durable plate, tags or label in accordance with MlL-P-15024, Plate shall be marked with the following information:

Sample for first article inspection
CONTAINERS SHIPPING & STORAGE Mk 300 Mod 0 and Mod 1
Plants where components are manufactured
Name of manufacturer or supplier
Contract or order number
Submitted by (Name and Date) for first article inspection in accordance with the requirements of MlL-C-81084A(OS) under (reference applicable contract number).

4.5.1 First article test equipment. The following items of test equipment are required to perform the first article tests specified herein.

a. A compressor or pressurized tank capable of pressurizing the container to 1.0 psig, minimum. The compressor or pressurized tank shall have a pressure gauge accurate to ±0.1 psig.

b. Vibration test equipment capable of applying and monitoring the magnitude and direction of vibration specified in 4.5.2.

c. A hoist or forklift truck capable of lifting and moving the container as specified in 4.5.3.

4.5.2 Vibration. Test the container to the vibration tests of 4.5.2.1 and 4.5.2.2 for cushioned items and as specified in MlL-STD-648. Evidence of damage to the contents or the inability of the container to pass the acceptance tests and examinations (see 4.7) constitutes a failure.

4.5.2.1 Variable frequency. The container shall withstand vibration of 5 to 60 Hz at 1 inch double amplitude or a multiple of gravitational acceleration (g's) of 1.3, whichever is less.

4.5.2.2 Natural frequency. The container shall withstand vibration at the resonant frequency of the packaged-article cushioning system at 1.3 g's or 1 inch double amplitude, whichever is less, for at least 30 minutes.

4.5.3 Rough handling. Immediately after the following tests (see 4.5.3.1 through 4.5.3.4) examine the container. Evidence of load shifting or binding as the contents are removed from the container constitute a failure.

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4.5.3.1 Cornerwise-drop (rotational). Test the container to the rotational drop test as specified in FED-STD-101, Method 5005 or MlL-STD-648.

4.5.3.2 Impact. Test the container to the impact test as specified in MlL-STD-648, except that the impact velocity shall be 7.6 feet per second.

4.5.3.3 Tip-over. Test the container to the tip-over test as specified in MlL-STD-648.

4.5.3.4 Skids. Test the container to the pushing and towing test as specified in MlL-STD-648 and FED-STD-101, Method 5011.

4.6 Periodic production tests. The periodic production sample, selected in accordance with 4.3.2, shall be subjected to the following tests (see 4.6.2). The contractor shall prepare and forward a test report in accordance with MlL-D-831 to the activity designated in the contract. The test report data shall consist of the following (see 6.3):

a. Items 4.5.1a through 4.5.1c
b. Test results of periodic production tests (4.6.2a through 4.6.2f)

4 6.1 Periodic production test equipment. The following items of test equipment are required to perform the periodic production tests specified herein.

a. A compressor or pressurized tank (see 4.5.1a)
b. Vibration test equipment (see 4.5. lb)
c. A hoist or forklift truck (see 4.5.1c)

4.6.2 Periodic production test schedule. Subject the container to the following tests in the order indicated:

a. Vibration (see 4.5.2)
b. Cornerwise-drip, rotational (see 4.5.3.1)
c. Impact (see 4.5.3.2)
d. Tip-over (see 4.5.3.3)
e. Skids (see 4.5.3.4)
f. Acceptance tests and examinations (see 4.7)

4.7 Acceptance tests and examinations. The following tests and examinations (see 4.7.2 and 4.7.3) shall be conducted on each container to be delivered. Failure to pass any test or examination specified herein shall be considered cause for rejection of the unit.

4.7.1 Acceptance test equipment. The only item of test equipment required to perform the acceptance tests specified herein is a compressor or pressurized tank (see 4.5.1a).

4.7.2 Airtightness. Remove breather valve (see Drawing 2452719) and seal opening. Pressurize the container to 0.75 + 0, - 0.25 psig and seal inlet. Allow the container to stand for 30 $\pm$ 5 minutes. A pressure drop in excess of 0.3 psig constitutes a failure.

4.7.3 Examination of product. The container shall be examined for conformance to the requirements specified herein and on drawings listed on LD541201 or DL2613831. The presence of any one of the following defects shall constitute a failure:

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a. Container not complete; component parts missing
b. Component parts not new; evidence of reclaimed, used, or salvaged components
c. Evidence of use of unauthorized repairs
d. Shock mounts not correctly installed
e. Rubber surfaces of saddles and straps missing, damaged, or deteriorated
f. Evidence of shock mount deterioration
g. Use of unauthorized materials
h. Evidence of substandard weld procedures
i. Loose mechanical connections
j. Paint damaged or coverage not complete
k. Identification plate and markings not attached or not located as specified; not legible; data incomplete
l. Packing not as specified in section 5

5. PACKAGING

5.1 Preservation and packaging. Not applicable.

5.2 Packing.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Not applicable.

5.2.3 Level C or industrial. Cleaning, drying, preservation, and packaging shall be accomplished in accordance with the contractor's own commercial practice when such practices have been approved by the contracting activity (see 6.5.3 and 6.5.4).

5.3 Marking. Unless otherwise specified, labeling and marking of all shipments shall be in accordance with MlL-STD-129. Any special marking required by the contract or order shall be provided. Marking shall be applied only to the packing material, as specified. Carload or truckload shipments do not require marking.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The container is intended as a reusable container for use in both foreign and domestic shipment and indeterminate storage of one Torpedo Main Assembly Mk 46 Mod 0, or one Propellant Actuated Gas Pressure Generator, Mk 8 Mod 0, loaded or unloaded.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DoDISS to be sited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2)

c. Requirements from associated detail specifications (see 3.1)

d. Whether a first article inspection is required (see 3.2)

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e. Materials and treatments of materials and parts required, used in manufacture of te containers (see 3.3).

f. Responsibility and suitable facilities for performance of inspections (see 4.1)

g. Assigned activity for first article and periodic tests (see 4.3.1)

h. All additional samples required shall be provided and inspected at the expense of the contractor, when such additional samples are required because of failure of any container to pass prescribed periodic production inspections (see 4.3.2.1)

i. Test conditions for container (see 4.4)

j. Required level of preservation, packaging, and packing (see section 5)

k. Number of containers required

l. Any exceptions to this specification

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only the essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (CDRL) (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.6	DI-T-2072	Reports, test	.

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Aquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3.1 Waiver. The data requirements of 6.3 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by that offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results (see 4.6), and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Definitions.

6.5.1 Humidity. The humidity is the relative humidity of the environment to which the test set is exposed, expressed as a percent.

6.5.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches from a position of rest to peak. The acceleration is given in multiple of gravitational acceleration. The frequency is given in cycles per second (Hz).

6.5.3 Level C. Minimum protection designated as Level C, will be used for protection of material under known favorable conditions. In general the following criteria determine the requirements for this level of protection.

a. Use or consumption of the item at the first destination
b. Shock, vibration, and static loading during the limited transportation cycle
c. Favorable warehouse environmental for a maximum of 18 months
d. Effects of environmental exposure during shipment and intransit delays
e. Stacking and supporting superimposed loads during shipment and temporary storage.

When level C is specified, it must reference applicable Public Laws (Code of Federal Regulations) or a specific Federal or Military Specification, Standard, or Instruction.

6.5.4 Industrial packaging. Industrial packaging will be acceptable for any level of protection whenever the technical design details of the package meet all conditions of the level of protection specified. Industrial packaging must provide the same level of protection against physical and environmental damage as the military package. It will be marked to the level it meets. Bulk type practices such as are used in inter- and intraplant shipments or shipments to jobbers are not acceptable, unless they are the usual trade practices for selected commodities.

6.6 Part number or national stock number (NSN).

Part number 2159356
National stock number 8140-00-747-9657

6.7 Subject term (key word listing).

Airtightness
Gas generator
Main assembly
Reusable
Simulated load
Shipment
Shock loading
Vibration

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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